

Soumyabrata Bhat

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PROFILE

CS student at Heritage Institute of Technology with hands-on experience building full-stack web applications and ML systems. Comfortable across the stack from PostgreSQL schemas and REST APIs to PyTorch models and React frontends. Enjoy working on problems that sit at the intersection of mathematics and practical software, and have independently explored agentic AI and zero-shot learning outside coursework.

EDUCATION

Heritage Institute of Technology <i>B.Tech. in Computer Science & Engineering CGPA: 8.18</i>	Kolkata, West Bengal 2023 – 2027
Asian International School <i>CBSE – Class XII (PCM with CS) Percentage: 81%</i>	Howrah, West Bengal 2023
DAV Public School <i>CBSE – Class X (Science, Math Std.) Percentage: 94%</i>	Bokaro, Jharkhand 2021

TECHNICAL SKILLS

Languages: Python, JavaScript, Java, C, C++, SQL, R
Web Development: MERN Stack, Next.js, REST APIs, Tailwind CSS
Databases: Relational (PostgreSQL, MySQL, Oracle SQL) & NoSQL (MongoDB)
AI / ML & Deep Learning: Scikit-Learn, TensorFlow, PyTorch, Hugging Face Transformers; CNNs, RNNs, Transformers
Data Science & Mathematics: NumPy, Pandas, Matplotlib; Probability, Statistics, Linear Algebra, Calculus
Dev Tools: Git, Docker, Linux, Postman
Algorithms & Problem Solving: Object-Oriented Programming (OOP), Data Structures and Algorithms (DSA), Design & Analysis of Algorithms (DAA)

PROJECTS

- EcoPulse** | *Next.js 15, TypeScript, Supabase, Tailwind CSS, Leaflet, Chart.js* [G](#) EcoPulse [↗](#)
- Built a full-stack collaborative platform connecting NGOs, volunteers, schools, and donors to address poverty, hunger, and education gaps, featuring real-time geolocated aid mapping and resource distribution tracking
 - Designed a PostgreSQL schema with Row-Level Security across 10 tables; implemented role-based auth, impact analytics dashboard, and a multi-language Learning Hub
- Image Classifier** | *Python, FastAPI, React, PyTorch, OpenAI CLIP* [G](#) Image Classifier [↗](#)
- Developed a zero-shot image classification system using OpenAI CLIP to identify images across 1,000 ImageNet categories without task-specific training
 - Built a FastAPI inference backend with React frontend; integrated Wikipedia API for contextual descriptions and delivered top-5 predictions with confidence scores in 2 to 5 seconds

ACHIEVEMENTS & ADDITIONAL SKILLS

- Agentic AI System:** Independently built a local agentic AI system capable of autonomous multi-step reasoning and task execution, integrating LLM inference with custom tooling pipelines
- Hackathon Qualifier:** Qualified for hackathons on three separate occasions, demonstrating consistent performance in time-constrained, competitive problem-solving environments
- Fine Arts Honours:** Achieved First Division with Honours in Fine Arts; proficient in 3D modelling and game development workflows
- Trilingual:** Fluent in English, Hindi, and Bengali with strong cross-cultural communication skills across academic and collaborative settings
- Soft Skills:** Demonstrated skills including leadership, project management, public relations, time management, and critical thinking across projects and extracurricular activities